assignment 4

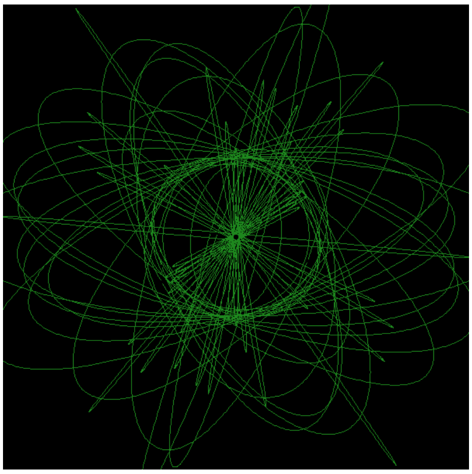
Status	Process
Description	take assignment 3 trained model and make it interactive in real-time with human input and robotic output
Completion	@January 26, 2026
Design Tools	

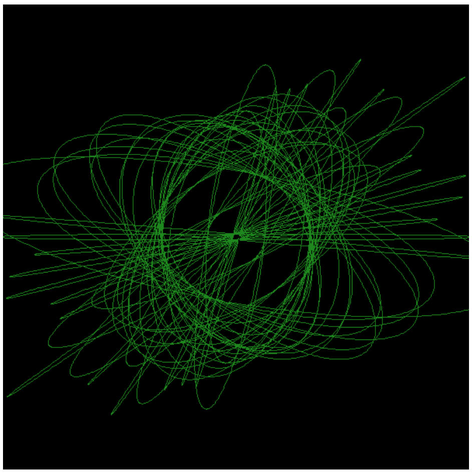
I chose to do an emotion detection assignment because i am interested in how humans can't always identify emotions and especially interested in our micro-expressions. Little and fast expressions that show on our face before we can think about it, usually completely invisible to humans. So this project is the beginning of experimenting with how a machine could pick up these micro-expressions.

For this assignment, I was more looking at how emotions could be objetifiable truths and what makes someone who is happy, look visually happy for example. I find it interesting how everyone makes the same facial movements. The robotic application I chose for emotion detecting relates to my mini project in which I would let people **'draw with their emotions'** instead of sound. Much like how we assimilate sounds to specific shapes like a low-pitched bass to a more rounded shape and high-pitched high hats to spikier shapes, I wanted to explore what shapes we would assimilate with emotions.

For my robotic application I would add a photo booth style interface for the user for them to take 4 pitcures, as well as a souvenir they could take away from the project, but mainly so that they would change their facial expressions and emotions. My project would cut this 20 second video clip into 50 frames, each frame saving the average emotion detected and confidence level. Each frame would correlate to an oval in a spirograph pattern, and each emotion and confidence score would correlate to a different shaped and sized oval. Creating a drawing that reflects the photo booth experience, in other words, the user's entire emotional journey and the machine's ability to read it. For example, anger = spiky and small, like the frustrating ball you get in your stomach and happiness would be large and rounder ovals, so the spikiness and size of the ovals would be mapped to the different emotions. And the confidence score would be reflected through the smoothness of the line.

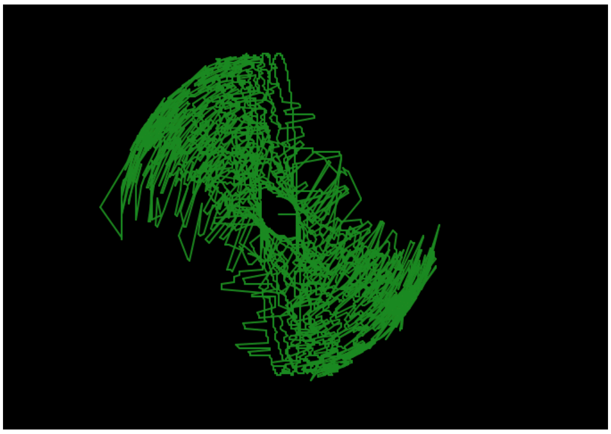
The images below are an example of the type of drawing it could create, though this is actually a drawing from my mini project with a similar idea but drawing with sound instead of emotions (though all these emotions would be 100% confident due to the smoothness of the line).





People can come away from their interaction with my project with a photo booth style picture of them and the drawing as a form of them drawing with their expressions and the machine trying to understand.

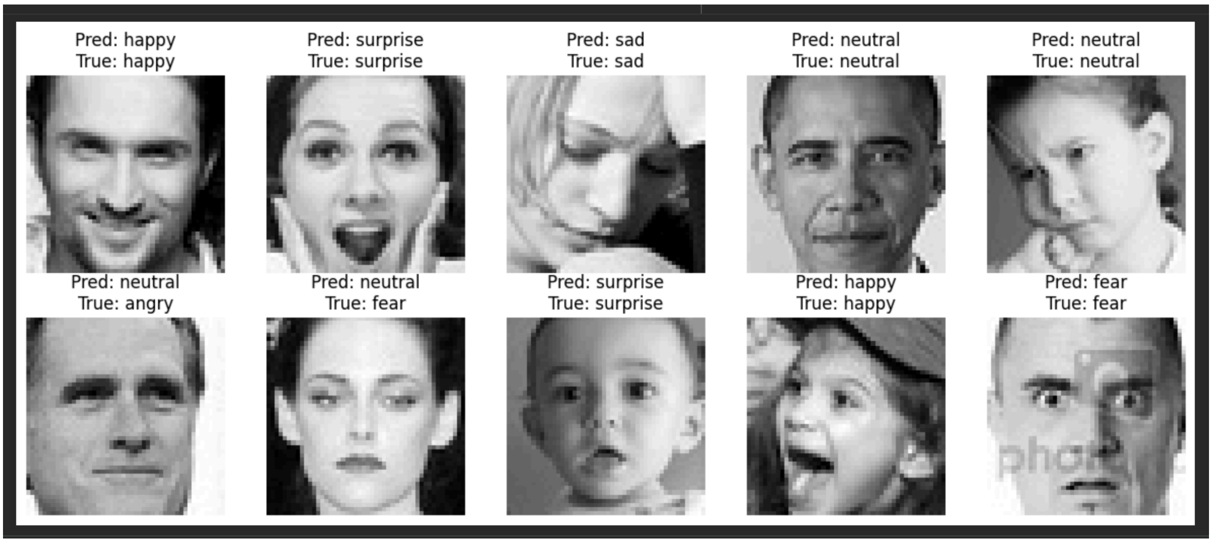
An unsure output (lower confidence scores) would look more like this:



SO I added my trained CNN emotion detecting model with Mediapipe's face detection model and made it read live emotions on detected faces with my webcam. I added the same transformations to the detected faces as I did to the dataset when training my model, making faces greyscale, 48,48 size, converted to tensors and normalised to match. I added smoothing with deque to calculate the average emotion within the last 5 predictions as it was detecting the emotions really fast. However, the model keeps getting stuck on 'surprised'. It can read 'happy' and 'anger' when really exaggerated, but it views neutral as surprised.

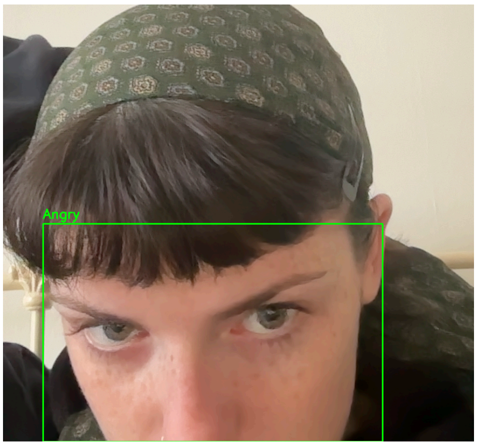
The averaging of the last 5 detected emotions could be strengthening the bias as if it's more likely to show surprised and I collect the last 5 detections, it's more likely that 3 of these detections will be surprised and the real emotion detected would get lost.

The bias could come from a number of things like the actual size of the dataset. The size of the images are 48,48 which is low-resolution which could easily be distorted when applying the transforms to a live camera. The lighting of the webcam could confuse the model as well. CNN focuses on edges and contrast and with surprised, there's lots of contrast due to open mouths and eyes, the lighting of the camera could be creating more contrast which the model has learnt to mean is surprised. The emotions detected were also trained with no background behind, being really up close images of faces and the bounding boxes I have added include the neck, hair and background, which could easily be a factor. Finally, the dataset has been proved to be wrong showing the true predictions as fear when in actuality, as a human, I can tell it's anger or sadness as neutral etc.



here i would say the predicted detections are more accurate than the true emotions it states.

However, this would just mean that my robotic drawing output would then be representing the process of the machine being unable to objectifiably detect emotions. It would be a drawing of the way a machine in no way can understand the nuance of emotions. It's difficult to even put emotions into 7 classes since both those faces (in the left bottom corner) shown above that it has predicted as neutral, even if still more accurate than angry and fear, is not neutral at all but probably pride and something that even I find hard to describe.



before adding confidence score, but you can see that it detects this as anger due to the angle and prominance of the eyebrows.

For my robotic concept I added a maximum of 50 frames every 5 detected frames to represent the 20s photo booth user interaction and each frame included the emotion detected and confidence score. I added the robotic output in the terminal: the shape and wobbleness of the pen. What would be outputted as a drawing like the spirographs shown above:

```
Drawing According to Emotions...
{'frame': 1, 'emotion': 'Happy', 'confidence': 0.86, 'shape': 'big slightly spikey oval', 'line_style': 'smooth line'}
{'frame': 2, 'emotion': 'Happy', 'confidence': 0.69, 'shape': 'big slightly spikey oval', 'line_style': 'moderately wobbly line'}
{'frame': 3, 'emotion': 'Happy', 'confidence': 0.49, 'shape': 'big slightly spikey oval', 'line_style': 'moderately wobbly line'}
{'frame': 4, 'emotion': 'Happy', 'confidence': 0.8, 'shape': 'big slightly spikey oval', 'line_style': 'smooth line'}
{'frame': 5, 'emotion': 'Happy', 'confidence': 0.66, 'shape': 'big slightly spikey oval', 'line_style': 'moderately wobbly line'}
{'frame': 6, 'emotion': 'Happy', 'confidence': 0.86, 'shape': 'big slightly spikey oval', 'line_style': 'smooth line'}
{'frame': 7, 'emotion': 'Happy', 'confidence': 0.74, 'shape': 'big slightly spikey oval', 'line_style': 'smooth line'}
{'frame': 8, 'emotion': 'Happy', 'confidence': 0.81, 'shape': 'big slightly spikey oval', 'line_style': 'smooth line'}
{'frame': 9, 'emotion': 'Happy', 'confidence': 0.59, 'shape': 'big slightly spikey oval', 'line_style': 'moderately wobbly line'}
{'frame': 10, 'emotion': 'Surprise', 'confidence': 0.48, 'shape': 'big spikey oval', 'line_style': 'moderately wobbly line'}
{'frame': 11, 'emotion': 'Surprise', 'confidence': 0.81, 'shape': 'big spikey oval', 'line_style': 'smooth line'}
{'frame': 12, 'emotion': 'Surprise', 'confidence': 0.47, 'shape': 'big spikey oval', 'line_style': 'moderately wobbly line'}
{'frame': 13, 'emotion': 'Angry', 'confidence': 0.55, 'shape': 'small spikey oval', 'line_style': 'moderately wobbly line'}
{'frame': 14, 'emotion': 'Surprise', 'confidence': 0.35, 'shape': 'big spikey oval', 'line_style': 'completely wobbly line'}
{'frame': 15, 'emotion': 'Surprise', 'confidence': 0.56, 'shape': 'big spikey oval', 'line_style': 'moderately wobbly line'}
{'frame': 16, 'emotion': 'Surprise', 'confidence': 0.33, 'shape': 'big spikey oval', 'line_style': 'completely wobbly line'}
{'frame': 17, 'emotion': 'Angry', 'confidence': 0.42, 'shape': 'small spikey oval', 'line_style': 'moderately wobbly line'}
{'frame': 18, 'emotion': 'Surprise', 'confidence': 0.34, 'shape': 'big spikey oval', 'line_style': 'completely wobbly line'}
{'frame': 19, 'emotion': 'Surprise', 'confidence': 0.52, 'shape': 'big spikey oval', 'line_style': 'moderately wobbly line'}
{'frame': 20, 'emotion': 'Surprise', 'confidence': 0.31, 'shape': 'big spikey oval', 'line_style': 'completely wobbly line'}
{'frame': 21, 'emotion': 'Surprise', 'confidence': 0.44, 'shape': 'big spikey oval', 'line_style': 'moderately wobbly line'}
{'frame': 22, 'emotion': 'Happy', 'confidence': 0.5, 'shape': 'big slightly spikey oval', 'line_style': 'moderately wobbly line'}
{'frame': 23, 'emotion': 'Happy', 'confidence': 0.68, 'shape': 'big slightly spikey oval', 'line_style': 'moderately wobbly line'}
{'frame': 24, 'emotion': 'Neutral', 'confidence': 0.29, 'shape': 'medium sized circle', 'line_style': 'completely wobbly line'}
{'frame': 25, 'emotion': 'Angry', 'confidence': 0.37, 'shape': 'small spikey oval', 'line_style': 'completely wobbly line'}
{'frame': 26, 'emotion': 'Angry', 'confidence': 0.28, 'shape': 'small spikey oval', 'line_style': 'completely wobbly line'}
{'frame': 27, 'emotion': 'Happy', 'confidence': 0.77, 'shape': 'big slightly spikey oval', 'line_style': 'smooth line'}
{'frame': 28, 'emotion': 'Happy', 'confidence': 0.81, 'shape': 'big slightly spikey oval', 'line_style': 'smooth line'}
{'frame': 29, 'emotion': 'Angry', 'confidence': 0.28, 'shape': 'small spikey oval', 'line_style': 'completely wobbly line'}
{'frame': 30, 'emotion': 'Surprise', 'confidence': 0.53, 'shape': 'big spikey oval', 'line_style': 'moderately wobbly line'}
{'frame': 31, 'emotion': 'Surprise', 'confidence': 0.53, 'shape': 'big spikey oval', 'line_style': 'moderately wobbly line'}
{'frame': 32, 'emotion': 'Neutral', 'confidence': 0.36, 'shape': 'medium sized circle', 'line_style': 'completely wobbly line'}
{'frame': 33, 'emotion': 'Surprise', 'confidence': 0.33, 'shape': 'big spikey oval', 'line_style': 'completely wobbly line'}
{'frame': 34, 'emotion': 'Sad', 'confidence': 0.45, 'shape': 'big round oval', 'line_style': 'moderately wobbly line'}
{'frame': 35, 'emotion': 'Sad', 'confidence': 0.43, 'shape': 'big round oval', 'line_style': 'moderately wobbly line'}
{'frame': 36, 'emotion': 'Sad', 'confidence': 0.48, 'shape': 'big round oval', 'line_style': 'moderately wobbly line'}
{'frame': 37, 'emotion': 'Surprise', 'confidence': 0.48, 'shape': 'big spikey oval', 'line_style': 'moderately wobbly line'}
{'frame': 38, 'emotion': 'Surprise', 'confidence': 0.68, 'shape': 'big spikey oval', 'line_style': 'moderately wobbly line'}
{'frame': 39, 'emotion': 'Surprise', 'confidence': 0.74, 'shape': 'big spikey oval', 'line_style': 'smooth line'}
{'frame': 40, 'emotion': 'Surprise', 'confidence': 0.65, 'shape': 'big spikey oval', 'line_style': 'moderately wobbly line'}
{'frame': 41, 'emotion': 'Sad', 'confidence': 0.56, 'shape': 'big round oval', 'line_style': 'moderately wobbly line'}
{'frame': 42, 'emotion': 'Sad', 'confidence': 0.65, 'shape': 'big round oval', 'line_style': 'moderately wobbly line'}
{'frame': 43, 'emotion': 'Surprise', 'confidence': 0.48, 'shape': 'big spikey oval', 'line_style': 'moderately wobbly line'}
{'frame': 44, 'emotion': 'Happy', 'confidence': 0.6, 'shape': 'big slightly spikey oval', 'line_style': 'moderately wobbly line'}
{'frame': 45, 'emotion': 'Happy', 'confidence': 0.7, 'shape': 'big slightly spikey oval', 'line_style': 'smooth line'}
{'frame': 46, 'emotion': 'Happy', 'confidence': 0.88, 'shape': 'big slightly spikey oval', 'line_style': 'smooth line'}
{'frame': 47, 'emotion': 'Happy', 'confidence': 0.89, 'shape': 'big slightly spikey oval', 'line_style': 'smooth line'}
{'frame': 48, 'emotion': 'Happy', 'confidence': 0.9, 'shape': 'big slightly spikey oval', 'line_style': 'smooth line'}
{'frame': 49, 'emotion': 'Happy', 'confidence': 0.87, 'shape': 'big slightly spikey oval', 'line_style': 'smooth line'}
{'frame': 50, 'emotion': 'Happy', 'confidence': 0.5, 'shape': 'big slightly spikey oval', 'line_style': 'moderately wobbly line'}
```